

Appendix

When Does Context Pay Off? TF-IDF, Word2Vec and BERT for Sentiment Classification under Label, Compute and Domain Constraints

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Repository: <https://github.com/jaffer1009/Text-Representation-Comparison>

1. References

1.1 Academic literature

- Bolukbasi, T., Chang, K.-W., Zou, J., Saligrama, V., & Kalai, A. (2016). Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings. *Advances in Neural Information Processing Systems (NeurIPS)* 29.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *Proceedings of NAACL-HLT 2019*, 4171–4186.
- Harris, Z. S. (1954). Distributional Structure. *Word*, 10(2–3), 146–162.
- Maas, A. L., Daly, R. E., Pham, P. T., Huang, D., Ng, A. Y., & Potts, C. (2011). Learning Word Vectors for Sentiment Analysis. *Proceedings of ACL-HLT 2011*, 142–150.
- Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient Estimation of Word Representations in Vector Space. *arXiv:1301.3781*.
- Pang, B., & Lee, L. (2005). Seeing Stars: Exploiting Class Relationships for Sentiment Categorization with Respect to Rating Scales. *Proceedings of ACL 2005*, 115–124. — source of the Rotten Tomatoes sentence corpus.
- Pennington, J., Socher, R., & Manning, C. D. (2014). GloVe: Global Vectors for Word Representation. *Proceedings of EMNLP 2014*, 1532–1543.
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *Proceedings of EMNLP-IJCNLP 2019*, 3982–3992. — establishes that an un-fine-tuned [CLS] vector is a poor sentence representation; directly motivates our dual-pooling design.
- Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv:1910.01108*.
- Sparck Jones, K. (1972). A Statistical Interpretation of Term Specificity and its Application in Retrieval. *Journal of Documentation*, 28(1), 11–21. — the origin of IDF weighting.
- Strubell, E., Ganesh, A., & McCallum, A. (2019). Energy and Policy Considerations for Deep Learning in NLP. *Proceedings of ACL 2019*, 3645–3650. — motivates the compute axis (H2).
- Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., & Bowman, S. R. (2019). GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding. *Proceedings of ICLR 2019*.

1.2 Datasets, models and software

Non-academic sources are cited here only in the categories the guidelines permit: dataset documentation, model cards and official software documentation.

Hugging Face. imdb dataset card. <https://huggingface.co/datasets/imdb> — distribution used; underlying data is Maas et al. (2011).

Hugging Face. rotten_tomatoes dataset card. https://huggingface.co/datasets/rotten_tomatoes — underlying data is Pang & Lee (2005).

Hugging Face. distilbert-base-uncased model card. <https://huggingface.co/distilbert-base-uncased>

Twitter US Airline Sentiment (CrowdFlower / Figure Eight, 2015). — crowd-annotated tweets; used out-of-domain only, binarised by dropping the neutral class and class-balanced by subsampling.

Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Řehůřek, R., & Sojka, P. (2010). Software Framework for Topic Modelling with Large Corpora. Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks, 45–50. — gensim.

Wolf, T. et al. (2020). Transformers: State-of-the-Art Natural Language Processing. Proceedings of EMNLP 2020: System Demonstrations, 38–45.

2. Reproducibility

All experiments are reproducible from the repository linked above. The corpus is encoded once per representation and cached, after which each experiment is a logistic-regression fit on those cached features.

Step	Command	Approx. runtime (CPU)
Verify wiring	<code>python -m src.prepare --smoke</code>	~1 min
Encode + cache	<code>python -m src.prepare</code>	~X h
H1	<code>python -m src.exp1_label_efficiency</code>	~X min
H2	<code>python -m src.exp2_cost</code>	~X min
H3	<code>python -m src.exp3_robustness</code>	~X min
Figures	<code>python -m src.figures</code>	< 1 min

Random seeds are fixed (config.SEED = 42; five subsample seeds 42–46) and recorded per row in the result CSVs. Every number on the poster traces to a committed CSV under results/.

Full condition grid

The poster reports three representation families; the complete five-condition grid is given here.

Condition	Specification	Dim
TF-IDF	1–2 grams, min_df=2, sublinear tf, 50k max features	~50k sparse
Word2Vec (in-domain)	skip-gram on the IMDB train pool, window 5, min_count=5, 5 epochs, mean-pooled	300
GloVe (pretrained)	glove-wiki-gigaword-300, mean-pooled	300
DistilBERT [CLS]	frozen, max_len=256, final-layer [CLS]	768
DistilBERT mean-pool	frozen, max_len=256, mask-aware mean of final layer	768

Classifier held constant across all conditions: LogisticRegression (max_iter=2000), with $C \in \{0.01, 0.1, 1, 10, 100\}$ selected on the 5,000-review development split. Dense features are standardised; sparse TF-IDF is left unscaled, as it is already L2-normalised and centring would destroy sparsity.

3. Use of Generative AI

Claude (Anthropic) was used for research and study purposes only.